Enterprise Asset Management (EAM)

Curriculum: Introduction to S/4HANA using Global Bike



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Teaching material - Information

Teaching material - Version

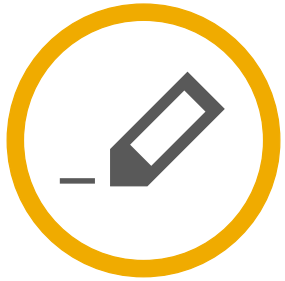
- 4.3 (July 2025)

- Software used
  - SAP S/4HANA 2023
  - Fiori 3.0
- Model
  - Global Bike

- Prerequisites
  - No Prerequisites needed

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Module Information



Authors

- Stefan Weidner
- Robert Häusler
- Chris Bernhardt
- Babett Ruß



Target Audience

- Beginner

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Module Information



Learning Objectives

You are able to

- name some functionalities of the EAM module.
- define the central organizational structures of the EAM module.
- summarize the master data which is most important for the EAM module.
- explain a standard process of the Enterprise Asset Management.

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Functionality

- Planned Repair
- Instant Repair
- External Assignment
- Refurbishment
- Preventive Plant Maintenance
- Project oriented Plant Maintenance
- Shift Reports and Shift Notes

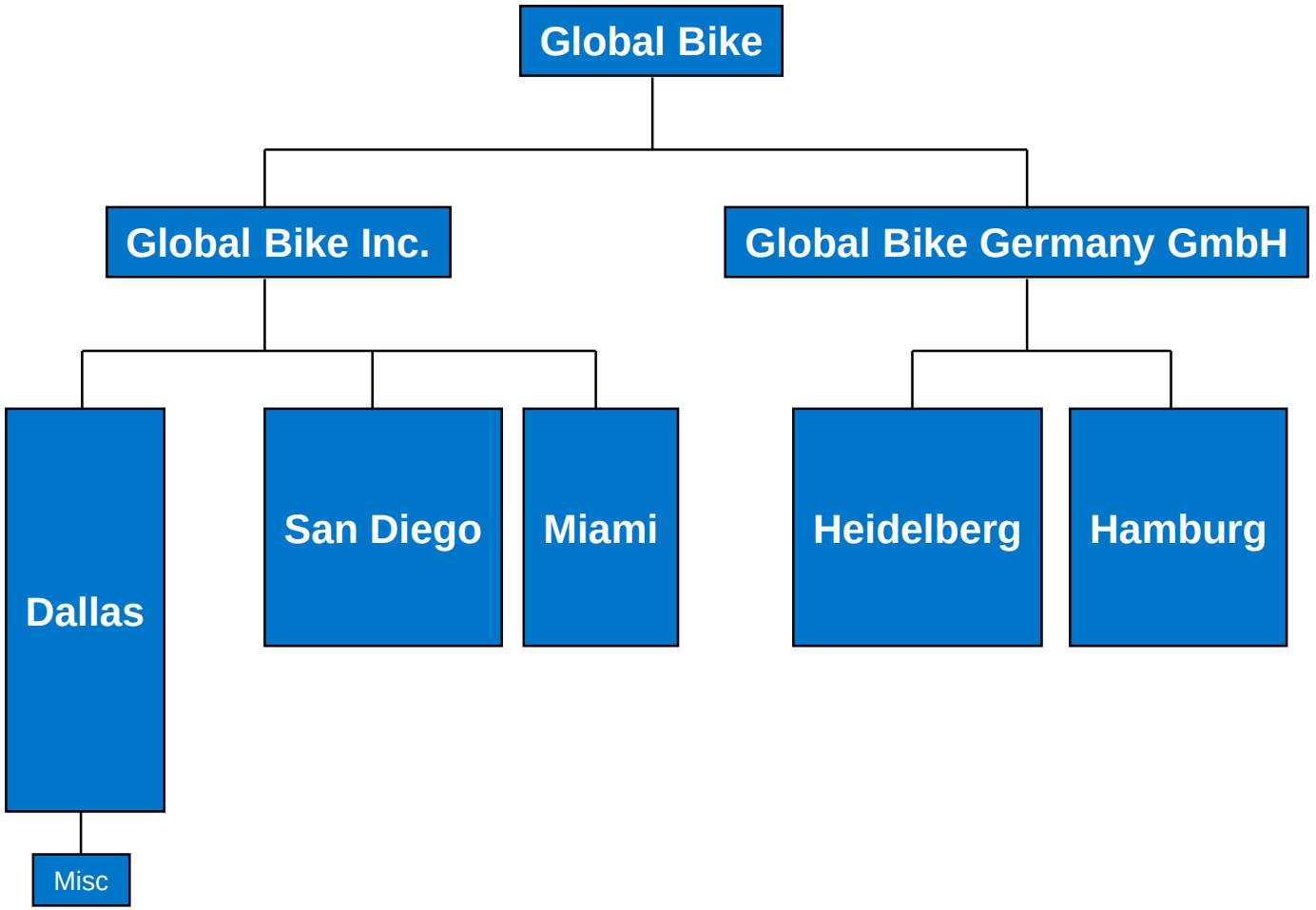
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Agenda

- EAM Organizational Structure
- EAM Master Data
- EAM Processes

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GBI Structure for Enterprise Asset Management



Client

Company Code

Plant

Maintenance Plant

Responsible Maintenance Planning Plant

Storage Location

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EAM Organizational Structure

- Client
  - An independent environment in the system
- Company Code
  - Smallest org unit for which you can maintain a legal set of books
- Plant
  - Operating area or branch within a company
    - Manufacturing, distribution, purchasing or maintenance facility
- Storage Location
  - An organizational unit allowing differentiation between the various stocks of material in a plant
  - Is required to store spare parts

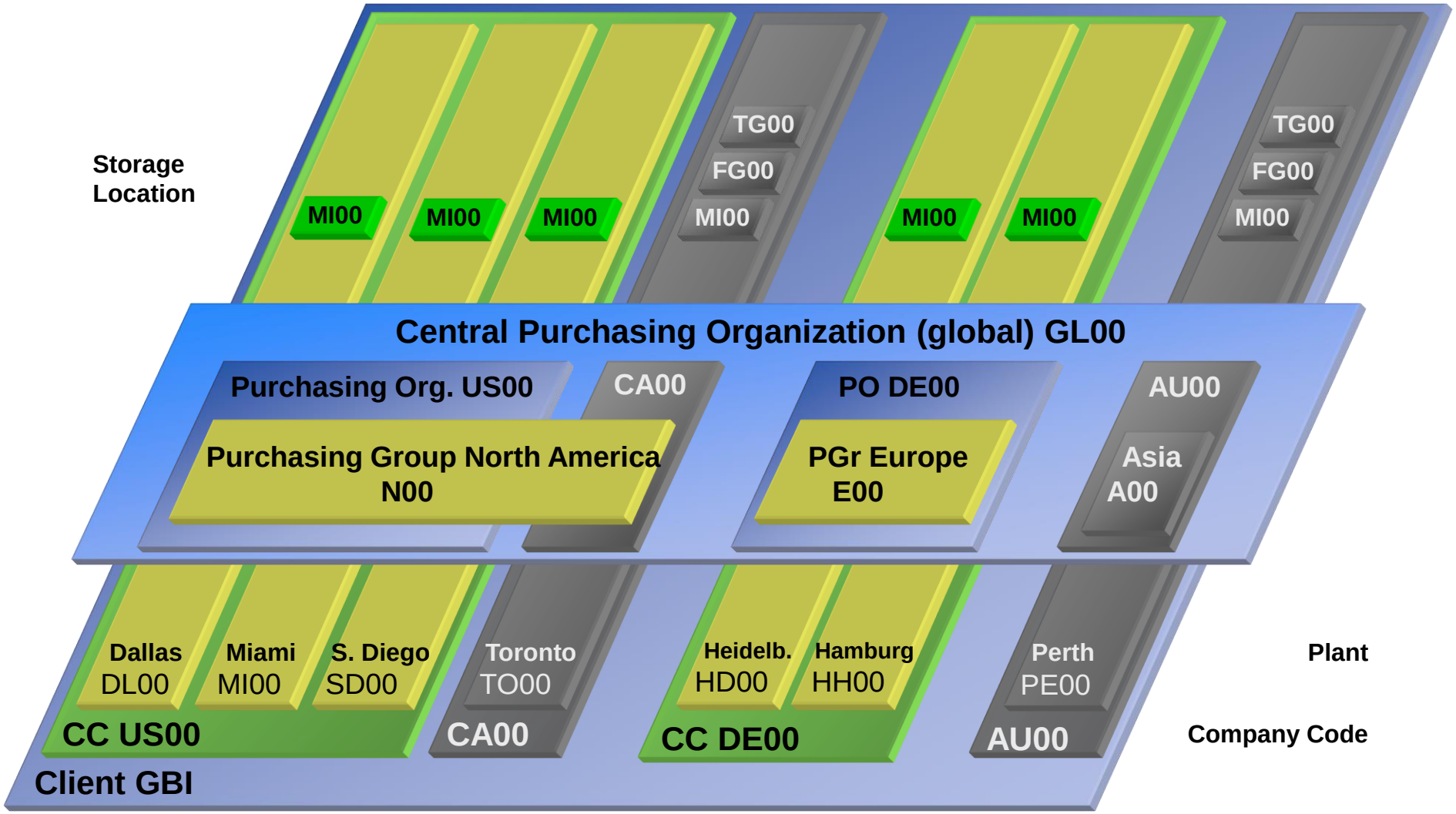
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EAM Organizational Structure

- Maintenance Plant
  - Plant in which the technical objects of a company are installed
- Maintenance Plant oriented organizational units:
  - Location (e.g. building number, coordinates)
  - Plant Section (responsibility for working assets)
  - Work Center (process measures of plant maintenance)
- Maintenance Planning Plant
  - Plant in which maintenance tasks are planned and prepared
- Planning Plant oriented organizational units :
  - Maintenance Planner Group (responsible for planning and processing maintenance tasks)
  - Work Center

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GBI Enterprise Structure in SAP ERP (EAM)



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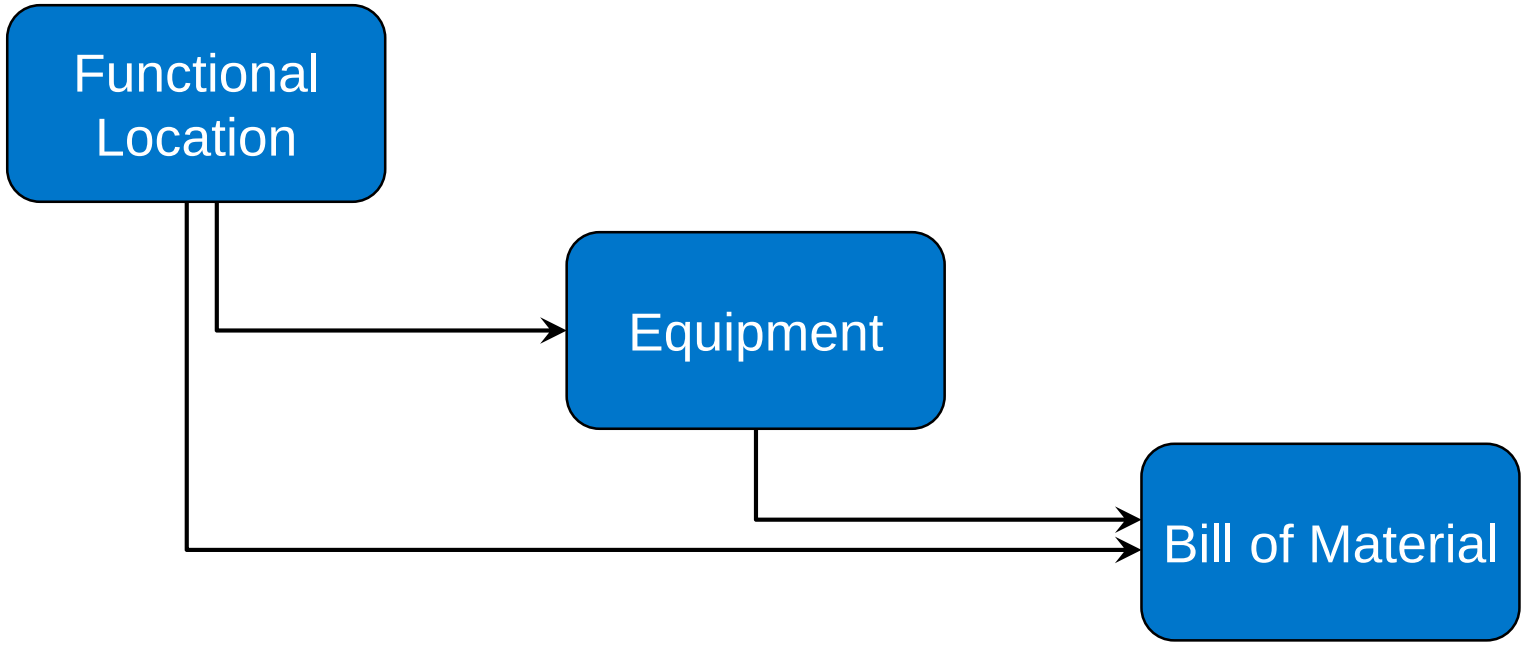
Agenda

- EAM Organizational Structure
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- EAM Processes

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EAM Master Data

- Functional Location – Master Data
- Equipment – Master Data
- Bill of Material



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Functional Location – Definition

- Hierarchically organized structure which represents e.g. a technical system, a building or a part of it
- Locations are normally fixed and do not move
- Structures maintenance objects according to following criteria:
  - spatial (e.g. buildings)
  - technical (e.g. pumping plant)
  - functional (e.g. production of bicycle frames)
- Functional locations can be used to sub-divide the objects into similar maintenance units
- Functional locations can contain different types of equipment

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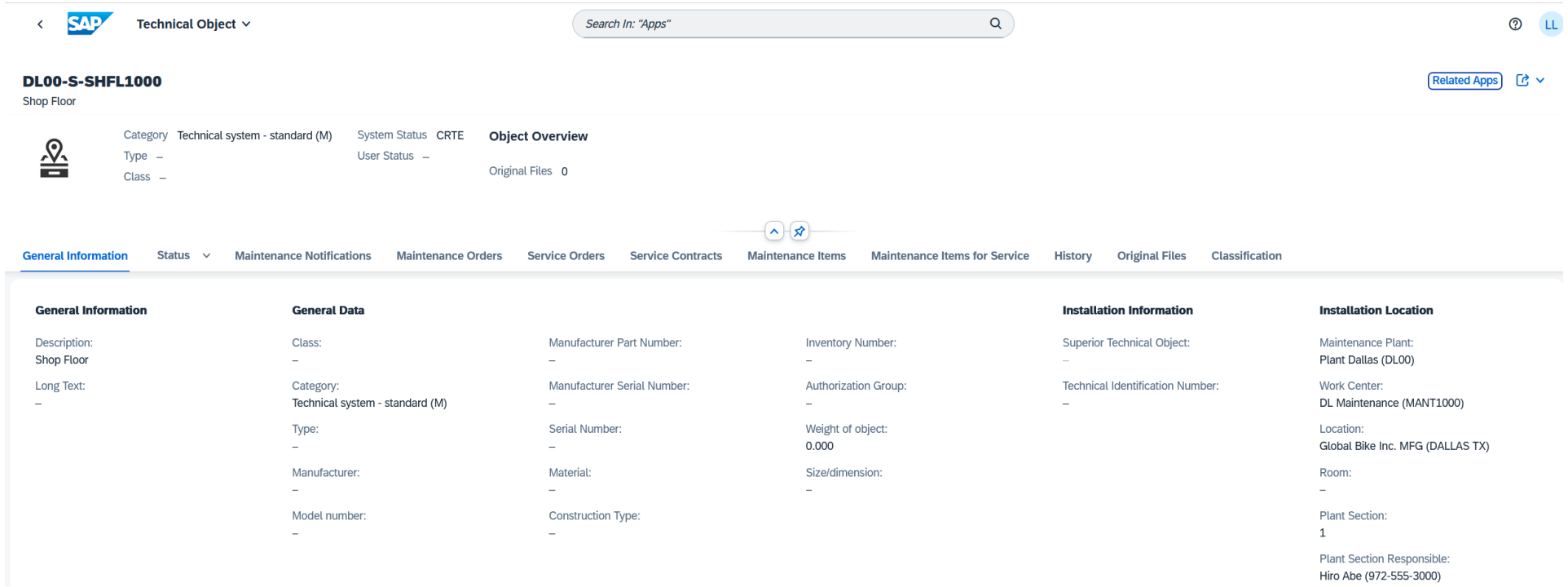
Functional Location – Advantages

- Structural depiction of a technical plant
  - Structure indicator makes hierarchy levels visible
- Planning and performing maintenance tasks
- Verification of maintenance tasks
- Data gathering for extended periods of time
- Cost monitoring
- Behavioral analysis individual areas under different conditions

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Functional Location – Master Record

- Functional Location Master Data
  - contains all information about the functional location
  - Could be required in order to perform maintenance activities



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Equipment – Definition

- Individual and autonomous technical unit
- Physical object for which maintenance tasks are planned and performed
- Mobile character
- Examples: pumps, personal computers, circulation fan, engines
- Equipments can be placed in functional locations

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Equipment – Advantages

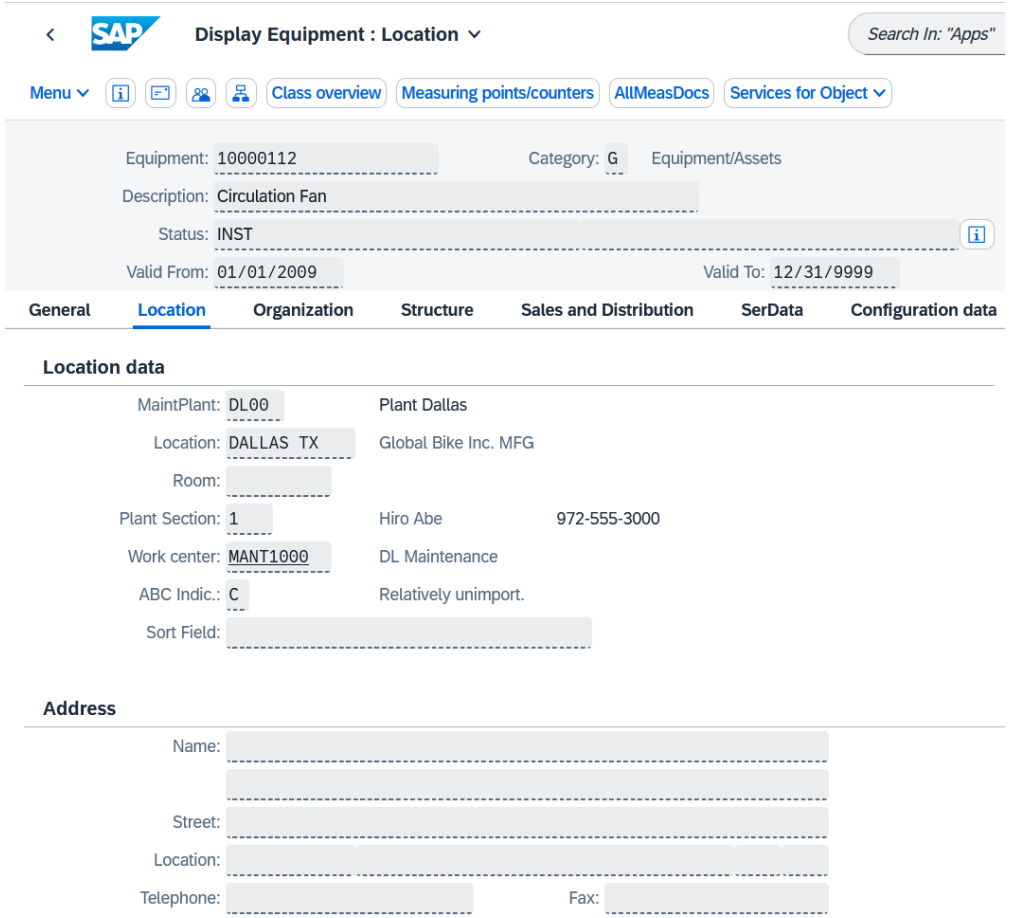
- Administration of individual objects and data
- Verification of maintenance tasks
- Object oriented cost monitoring
- Data gathering and analysis for extended periods of time

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Equipment – Master Record

- Equipment Master Data
  - contains all information around equipment

- Information in four views:
  - General
  - Location
  - Organization
  - Structure



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Bill of Material – Definition

- Complete formally structured list of all components
- Material BOMs can be allocated to functional locations or equipment
- Three types of BOMs in Enterprise Asset Management:
  - Material BOMs
  - Equipment BOMs
  - Functional location BOMs

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Bill of Material – Advantages

- Structuring of objects
- Service parts planning in maintenance orders and maintenance task lists
- Within plant maintenance BOMs are used if there are similar objects that can be maintained

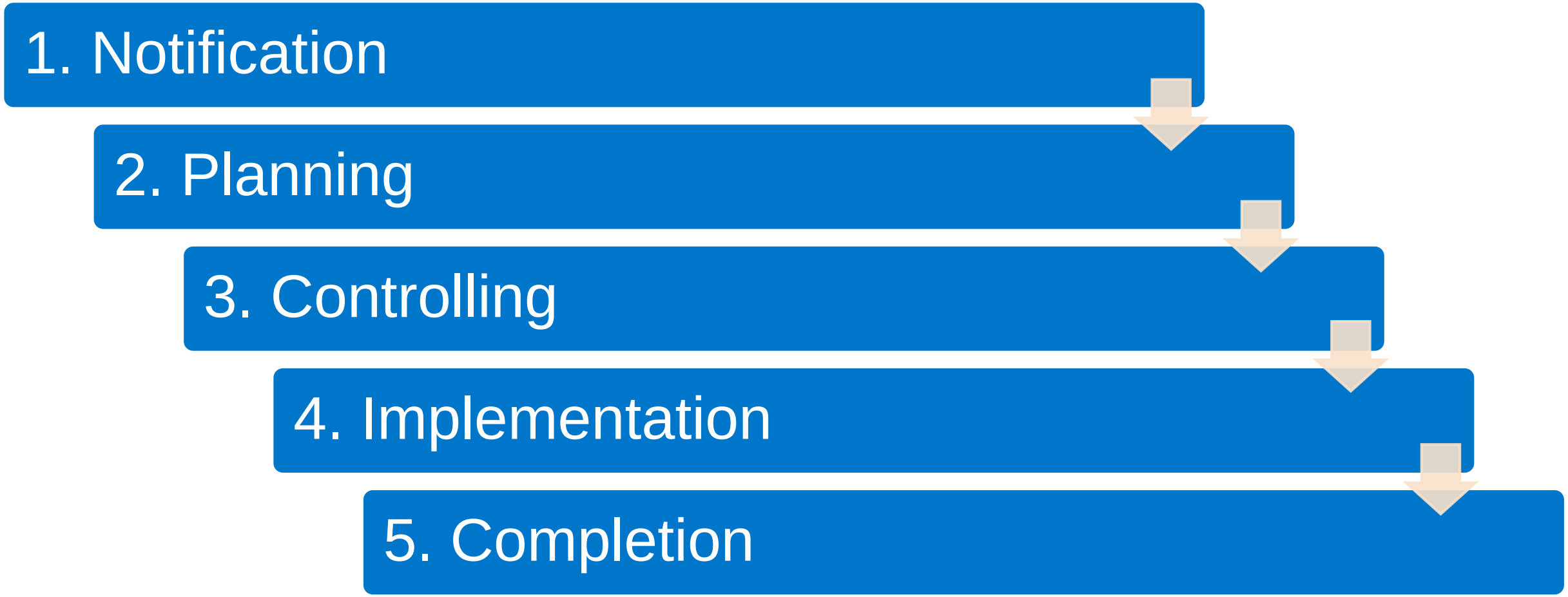
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Agenda

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EAM Process (Planned Repair)

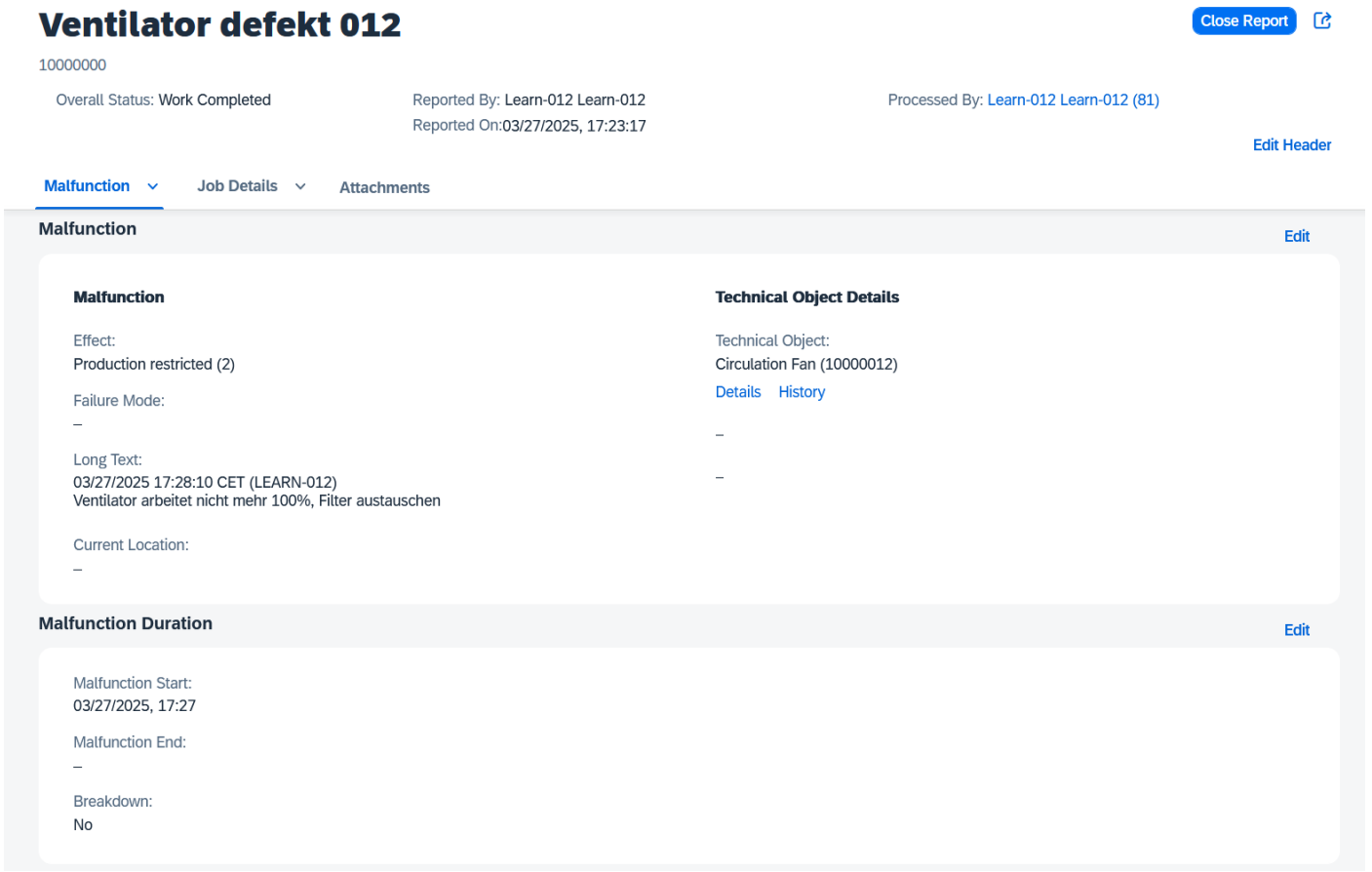


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Notification

- Malfunctions and other requirements are gathered in notifications

- Content:
  - Technical object
  - Reported by
  - Effect
  - Description
  - Date
  - Breakdown



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Notification

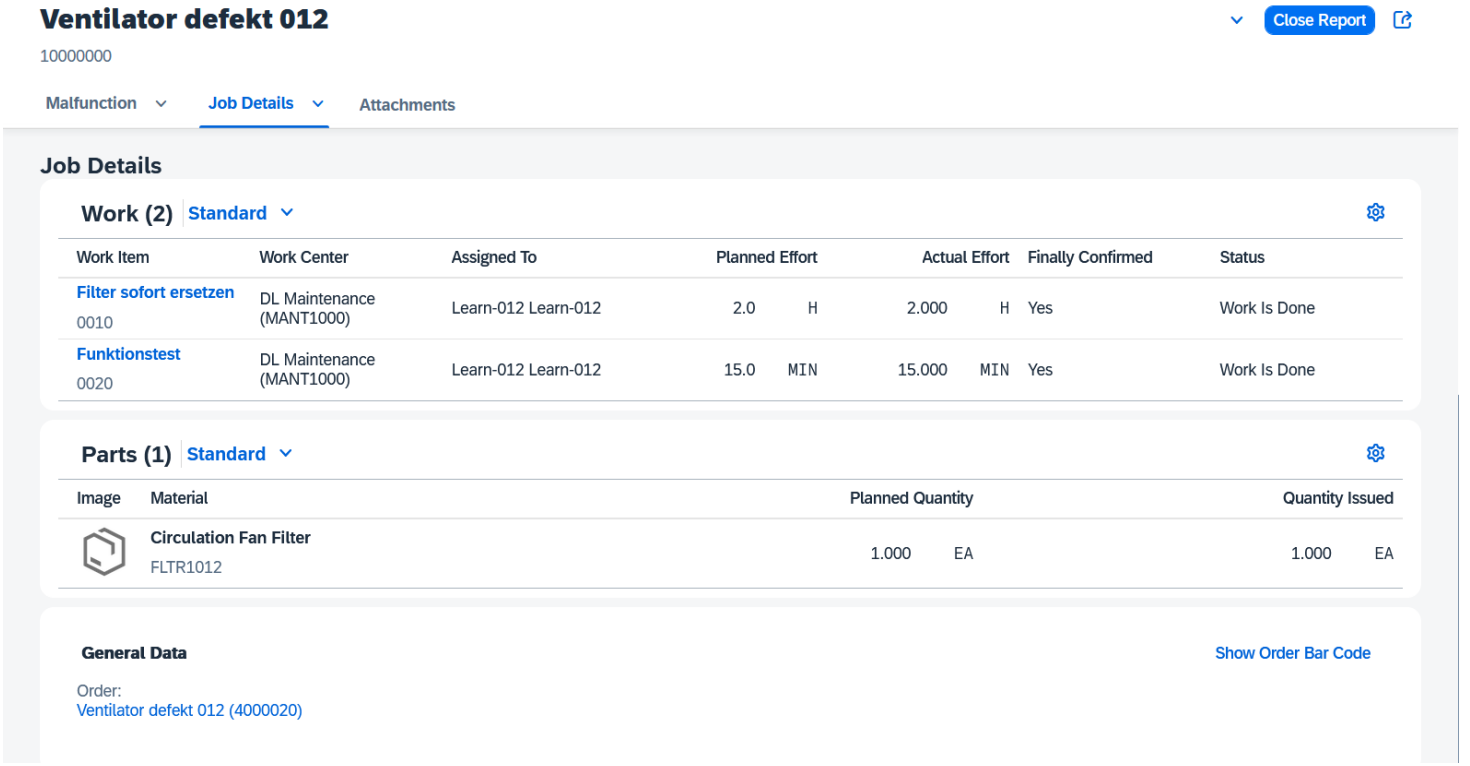
- Notifications are used to tell maintenance that something needs fixing
- They may or may not contain technical information
- They will often highlight a breakdown that needs to be actioned quickly
- Requirement of necessary maintenance measures
- Notifications can also be used to record works already carried out because a history needs to be maintained

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Planning

- Order creation and order planning is normally triggered from a notification

- Order planning can include:
  - The steps to be Performed (Work)
  - Required spare parts (Part)



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Planning

- Maintenance orders can be created from maintenance plans, notifications or created without any reference document
- Responsibilities for external or internal work are defined
- Material reservations take place order upon release
- Production resources like protective clothing and specialist tooling can be listed
- Planned costs are calculated based on internal activity rates and external service costs

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Maintenance Order Control Functions

- Control functions for maintenance orders can vary depending on the type of order
- Order release is carried out once planning has been completed
- Maintenance orders provide the following functions:
  - Mass change or mass editing
  - Availability check for spare parts
  - Capacity requirements planning
  - Printing work instruction papers

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Execution and Completion

- Order Execution
- Planned and unplanned material withdrawals (goods issue) are possible
- Order Completion Steps:
  - Time confirmation
  - Technical completion confirmation
  - Technical completion
- Customer billing can be performed after successful technical completion

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